

Gurumanie Singh Dhiman

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Seeking an internship or co-op in a cybersecurity-focused environment, specializing in networking, infrastructure, and backend, starting **Summer 2026**. Passionate about designing secure, scalable systems and integrating software with networks and security.

Education

Master of Science, Cybersecurity (Concurrent) **Expected May 2027**

Bachelor of Science, Computer Engineering (GPA: 3.86/4) **Expected May 2026**

Iowa State University, Ames, IA

- **Certifications:** CompTIA Security+ (SYO-701) certified, CCNA (in progress — expected 12/2025)
- **Competitions:** Placed 3rd in Cyber Defence Challenge, advancing to Internationals in Spring 2026
- **Relevant Coursework:** Network Protocols and Security, Information System Security, Computer and Network Forensics

Technical Skills

- **Security:** PKI, TLS, IAM, AAA, IDS/IPS, logging/monitoring, incident response
- **Networking:** Switching/Routing, VLAN, OSPF, DHCP/DNS, RF/interference, Wireshark
- **Tooling/Infra:** GitLab CI/CD, Docker, Linux (systemd, journalctl), LibreNMS, OpenStack, Git
- **Software Development Process:** Agile, Scrum, SDLC, Risk Management
- **Programming:** Java, C, Python, Bash

Experience

Wireless Network Systems Engineer, ARA Wireless Living Lab, Iowa State University **Jun 2025 – Current**

- Developed Python and Bash automation and GitLab CI/CD pipelines to streamline deployment of network updates, reducing manual intervention and cut deployment time from 2 hrs to 30 min.
- Containerized services with Docker and managed processes with systemd/journalctl, created automated health checks and restart policies to improve uptime for the systems.
- Improved Flask-based web applications to support functionality and demonstrate secure, scalable system integration.
- Wrote Sphinx-based technical documentation and runbooks to standardize deployments and shorten onboarding.

SI Leader for Data Structures and Algorithms, Iowa State University **Aug 2024 – Dec 2024**

- Led weekly sessions for 80+ students; created materials and practice problems focused on algorithmic complexity, debugging, and efficient implementation, maintained an attendance retention of 90%.

Projects

Secure Wireless Aircraft Data Load, Iowa State University **Aug 2025 – Current**

- Engineering a secure wireless data transfer architecture using Raspberry Pis, a dedicated hardware WAP, and a VM-hosted RADIUS server implementing PKI/TLS-based mutual authentication.
- Applying cryptographic key management, digital certificates, and TLS protocols to ensure data confidentiality and integrity when transmitting data across the wireless channel.
- Performed threat modeling and vulnerability analysis to enhance our system's resilience against attacks.
- Evaluated compliance with airline cybersecurity policies and demonstrated a 20% reduction in operator workload through automation and secure data handling.

Applied Cybersecurity Labs, Iowa State University **Jan 2025 – Current**

- Used OpenSSL and scripted tooling to decrypt exfiltrated files and extract hidden keys.
- Cracked /etc/shadow hashes with custom Python/Bash scripts against a 100k-password corpus and validated VM access.
- Performed slowloris DoS tests in isolated networks and implemented iptables rules to harden services.